MAHATMAGANDHIUNIVERSITY,KOTTAYAM

MGU-UGP (HONOURS)
THIRDSEMESTEREXAMINATION
(2024ADMISIONONWARDS)

MG3DSEECT204 SMARTSENSORSWITHIoT

Duration:2 hours MaximumMarks:70

**Remember(K),Understand(U),Apply(A),Analyze(An),Evaluate(E),Create(C),Skill(S),Interest(I) and Appreciation (Ap)*

PartA

MultipleChoiceQuestions
AnswerAllQuestions
EachquestioncarriesOnemark

1	Whichsensorworksontheprincipleofchangeinresistanceduetolight intensity? (a)LDR(b)Thermistor(c) StrainGauge(d) Phototransistor	1	U
2	Halleffectsensorsareprimarilyused for: (a)Temperaturemeasurement(b)Currentsensing(c)Pressuremeasurement(d) Ultrasonic sensing	1	U
3	Whichofthe followingisapassivetransducer? (a)Thermocouple(b) LVDT(c) Photodiode(d)All oftheabove	2	U
4	Themain advantageof piezoelectricsensorsis: (a)Highfrequencyresponse (b)Low cost (c)Easy calibration (d)Noneoftheabove	1	U
5	WhichcomponentisusedintheR-2Rladder network? (a)Resistors(b)Capacitors(c)Inductors(d) Diodes	2	U
6	AWheatstonebridgeiscommonlyused with: (a)Straingauges(b)Hallsensors(c)LDRs(d)Ultrasonic sensors	2	U
7	Whichofthe followingisanactivetransducer? (a)Thermocouple(b)StrainGauge(c)Potentiometer(d) LDR	2	U
8	Theworkingprincipleof ultrasonicsensorsisbasedon: (a)Infraredlight(b)Soundwaves(c)Magneticfield(d)Capacitance change	1	U
9	Inindustrialautomation, proximitysensorsareusedfor: (a)Detectingobjectswithoutcontact(b)Measuring temperature (c)Sensingpressure (d)Allof the above	1	U
10	WhichIoTcloudservice ismentionedforuploadingsensordata? (a)AWS IoTCore(b)ThingSpeak(c)AzureIoTHub(d)IBMWatsonIoT	4	U
11	Thedeviceusedto convertdigital signalsto analog is: (a)ADC(b)DAC(c)Amplifier(d)Transducer	2	U
12	Inaservomotor,thepositioniscontrolled by: (a)Voltagefeedback(b)PWMsignal(c)Magneticflux(d)Opticalsensor	3	An
13	Whichactuatorproduces linearmotion? (a)Steppermotor(b)Linear actuator(c)Servo motor(d) Relay	3	An

14	Whichactuatorprovidesstepwiserotationalmovementwithhighpositional accuracy? a)Steppermotor b)Linearactuator c)Solenoid d)Relay	4	U
15	Relaysaremainlyusedincontrolsystemsfor: a)Speedcontrolofmotorsb) Poweramplificationc)ConvertingACto DC d)Switchinghighpowercircuitsusinglowpower signals	4	U
16	Whichofthefollowingactuatorsconvertsrotationalmotionintolinear motion? a)Servo Motor b) Stepper Motorc) Linear Actuator d) Relay	3	An
17	Whichactuatoriscommonly usedforprecise angularpositioncontrol in robotics? a)DCmotorb)Servo motorc)Linear actuatord)Solenoid	4	U
18	Whichofthefollowing isanAnalyse-level activity? (a)Definingasensor (b)Designing anactuatordriver circuit (c)Listingtypesof transducers (d)Statinglawsof thermocouples	3	An
19	Theterm'Industry2.0'refersto: (a)Firstindustrialrevolution (b)Secondindustrial revolution (c)ModernIoTera (d)Automated roboticsage	4	U
20	Asteppermotormovesin: (a)Continuousrotation(b)Discretesteps(c)Vibrationcycles(d)Fluidmotion	3	An
		1 X20 = 20	

PartB

ShortAnswerQuestions

Answerany**6**Questions

Eachquestioncarries**5marks**

21	Explaintheworkingprincipleandapplicationsof Halleffectsensors.	1	U
22	Describetheconstructionandoperationofa Wheatstonebridgewithastrain gauge.	2	U
23	Differentiatebetweenpassiveandactivetransducerswithexamples.	2	U
24	Explaintheinterfacingofan steppermotor.	3	An
25	DescribethestepstouploadsensordatatoThingSpeakusingESP32.	4	U
26	Compareservomotorsandsteppermotorsintermsofoperationand applications.	3	An
27	Explaintheprincipleofoperation ofanultrasonic distancesensor.	1	U
28	DescribeIndustry2.0 anditsimpacton automation.	4	U
		5 X6 = 30	

PartC

Essay

QuestionsAnswerany**2**Q

uestions

Eachquestioncarries**10marks**

29	Designandexplainadrivercircuitforcontrollingalinearactuatorusingan IoT-based system. Include block diagrams and working principles.	3	An
30	Discussindetailthedifferenttypesoftemperaturesensorsandtheirindustrial applications.	1	U
31	Explainthearchitectureofanindustrialautomationsystem.Includetheroleof smart sensors and IoT.	4	U
32	Analyzetheworkingof R2RADC withsuitableexamples.	2	U
		10 X2 = 20	